

Diversity-Guided Adversarial Reinforcement Learning with Expert Routing and Recovery for Robust Autonomous Driving

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ABSTRACT

Robust autonomous driving requires reinforcement learning policies that can remain reliable under vehicle-dynamics variations, adversarial interactions, and distribution shifts. Existing adversarial reinforcement learning methods often rely on a single adversary or a statically assigned policy population, which may produce redundant perturbations, weak state adaptation, and unstable training. To address these limitations, this paper proposes a diversity-enhanced adversarial reinforcement learning framework with three coordinated components. First, a unified diversity measure constructs a multi-modal kernel from payoff and behavioral information and evaluates population diversity through its eigenvalue spectrum. This encourages adversarial policies that are both strategically effective and behaviorally distinct. Second, a state-conditioned mixture-of-experts architecture dynamically selects and combines adversarial heads according to the current traffic state, allowing different experts to specialize in different driving situations. Third, a joint recovery mechanism combines checkpoint rollback with recovery-cell restart to restore reliable policy parameters and revisit informative intermediate states after persistent performance degradation. A progressive action-mixing policy is also introduced to increase adversarial influence gradually during training. Experiments are conducted in MetaDrive under six train--test configurations with different vehicle-dynamics distributions. The results show that the proposed framework improves episodic return, safety-related performance, robustness, and training stability compared with representative reinforcement learning and adversarial training methods. Ablation and sensitivity studies further confirm the complementary effects of diversity regulation, state-dependent expert routing, and joint recovery.